

N-Queens Problem Using Python

Introduction

The N-Queens problem is a well-known problem in computer science where N queens must be placed on an $N \times N$ chessboard so that no two queens attack each other. This means queens cannot share the same row, column, or diagonal. The program uses a backtracking approach to find all possible solutions.

Functions Used

`is_safe()`: Checks if a queen can be placed at a specific position without conflict.

`solve_n_queens()`: Uses backtracking to place queens column by column and explore solutions.

`n_queens()`: Main function that creates the board, calls the solver, and prints the solutions.

Conclusion

This program shows how the backtracking algorithm can solve the N-Queens puzzle by trying different placements and undoing moves when conflicts occur.